

SIMATS ENGINEERING (SAVEETHA SCHOOL OF
ENGINEERING) Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – Application-Based Questions (3 Questions)
Course Code: ITA0609 Course Title: Machine Learning
CO Assessed: CO1 – Analyze (BL 3)
40% of CIA

Assessment: Assessment Tool 1 –
Application Based Questions.
30 (3 Questions × 10 Mark) Weightage: Total Marks:
CO1 Statement: To acquire a foundational understanding of key machine learning concepts
including version spaces, candidate elimination, inductive bias, and heuristic
search (BL1)

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Section / Batch: SLOT A Date: 13.7.26

Code of Conduct
I (SESHRAM D / 192412012) certify that this submission is my original work and that I have
adhered to the guidelines
specified for this assessment. I understand that any violation of academic integrity rules will
result in disciplinary
action.

Signature of the Student: Seshram D
Instructions.

- This test contains 5 Short questions, each carrying 10 mark.
- All questions are application-based and require analytical thinking and practical implementation. · No negative marking.
Unanswered questions carry zero marks.
- Provide suitable algorithms, logic, calculations, or case-based solutions wherever required.

1) A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

```
import fractions

# --- Car counts in each section ---
section_a_cars = {'red': 5, 'blue': 3, 'black': 2}
section_b_cars = {'red': 4, 'blue': 6, 'black': 5}
section_c_cars = {'red': 7, 'blue': 2, 'black': 6}

total_a = sum(section_a_cars.values())
total_b = sum(section_b_cars.values())
total_c = sum(section_c_cars.values())

# --- Prior probabilities of choosing a section ---
# P(B) = 2 * P(A) and P(B) = 2 * P(C), so P(A) = P(C)
# Let P(A) = x, then P(C) = x, P(B) = 2x
# x + 2x + x = 1 => 4x = 1 => x = 1/4
P_A = fractions.Fraction(1, 4)
P_B = fractions.Fraction(2, 4) # or 1/2
P_C = fractions.Fraction(1, 4)

# --- Conditional probabilities of drawing a blue car from each section ---
P_Blue_given_A = fractions.Fraction(section_a_cars['blue'], total_a)
P_Blue_given_B = fractions.Fraction(section_b_cars['blue'], total_b)
P_Blue_given_C = fractions.Fraction(section_c_cars['blue'], total_c)

# --- Probability of drawing a blue car (P(Blue)) using Law of Total Probability
P_Blue = (P_Blue_given_A * P_A) + (P_Blue_given_B * P_B) + (P_Blue_given_C * P_C)

# --- Apply Bayes' Theorem: P(C|Blue) = [P(Blue|C) * P(C)] / P(Blue) ---
P_C_given_Blue = (P_Blue_given_C * P_C) / P_Blue

print(f"Probability of choosing Section A: {P_A}")
print(f"Probability of choosing Section B: {P_B}")
print(f"Probability of choosing Section C: {P_C}")
```

```
print(f"\nProbability of blue car given Section A: {P_Blue_given_A}")
print(f"Probability of blue car given Section B: {P_Blue_given_B}")
print(f"Probability of blue car given Section C: {P_Blue_given_C}")
print(f"\nOverall probability of drawing a blue car (P(Blue)): {P_Blue}")
print(f"\nThe probability that the car came from Section C, given it is blue,
```

Probability of choosing Section A: 1/4
Probability of choosing Section B: 1/2
Probability of choosing Section C: 1/4

Probability of blue car given Section A: 3/10
Probability of blue car given Section B: 2/5
Probability of blue car given Section C: 2/15

Overall probability of drawing a blue car (P(Blue)): 37/120

The probability that the car came from Section C, given it is blue, is: 4/37

2. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

1. (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
2. (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
3. (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
4. (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
5. (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
6. (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
7. (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
8. (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2
9. i) Identify the epoch at which the model begins to overfit.
10. ii) Suggest two effective techniques to reduce overfitting in this model.
11. iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Epoch	Training Loss	Validation Loss	Observation
1	2.8	2.4	Both losses decreasing
2	2.5	2.2	Model improving
3	2.2	2.0	Best validation performance
4	2.0	2.1	Validation loss starts increasing
5	1.8	2.3	Overfitting increases
6	1.6	2.6	Severe overfitting
7	1.5	2.9	Validation performance worsens
8	1.3	3.2	Strong overfitting

i) Identify the epoch at which the model begins to overfit.

The model **begins to overfit at Epoch 4**.

Reason:

- The **training loss continues to decrease** ($2.2 \rightarrow 2.0$), indicating the model is learning the training data.
- However, the **validation loss starts increasing** ($2.0 \rightarrow 2.1$), showing that the model's performance on unseen data is getting worse.

ii) Suggest two effective techniques to reduce overfitting.

1. Early Stopping

- Stop training when the validation loss starts increasing (around Epoch 3 or 4).
- This prevents the model from memorizing the training data.

2. Regularization (L1/L2 or Dropout)

- Apply L1/L2 regularization to penalize large weights, or use Dropout to randomly deactivate neurons during training.
- This helps the model learn more general patterns instead of memorizing the training data.

iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Overfitting occurs when the model learns the training data too closely, including its noise and unnecessary details.

Effects:

- Achieves **low training loss** but **high validation/test loss**.
- Performs well on the training dataset but poorly on new, unseen data.
- Reduces the model's ability to generalize, leading to inaccurate predictions in real-world applications.

Conclusion:

The model starts overfitting at **Epoch 4**, so training should ideally stop around **Epoch 3**, or techniques like **early stopping** and **regularization/dropout** should be applied to improve generalization.

3.Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

12. (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%

13. (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%

14. (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%

15. (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%

16. (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%

17. (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%

18. (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%

19. (h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%

20. i) Identify the epoch at which the model starts overfitting.

21. ii) Explain why validation accuracy decreases despite training

accuracy increasing. 22. iii) Suggest two methods to improve validation performance.

i) Identify the epoch at which the model starts overfitting.

The model **starts overfitting at Epoch 6**.

Reason: Validation accuracy reaches its highest value (**73%**) at **Epoch 5** and then starts decreasing from **Epoch 6**, while training accuracy continues to increase.

ii) Explain why validation accuracy decreases despite training accuracy increasing.

Validation accuracy decreases because the model begins to **overfit** the training data. It memorizes the training examples and their noise instead of learning general patterns. As a result, it performs better on the training data but worse on unseen validation data.

iii) Suggest two methods to improve validation performance.

1. **Early Stopping** – Stop training when the validation accuracy stops improving (around Epoch 5).
2. **Regularization (L2 Regularization or Dropout)** – Reduces overfitting by preventing the model from becoming too complex and improving its ability to generalize to new data.

